

PROJECT SYNOPSIS PRESENTATION

ResQride: A QR-Code Based Vehicle Communication Platform with Automated Crash Detection

GROUP NO:	

GUIDE : Mrs. R.M. Yemul

ResQride: A QR-Code Based Vehicle Communication Platform with Automated Crash Detection

TEAM DETAILS

Name Of The Student	Roll No.
1. Vedant Bora	2305111027
2. Jayesh Durugkar	2305111102
3. Abhijeet Garbhe	2305111008
4. Om Kirtankar	2305111080

1. INTRODCUTION

- Road accidents lose critical time in the “golden hour” due to delayed victim identification and contact.
- Existing options: costly OEM connected-car systems, or unreliable manual bystander action.
- Modern smartphones already carry GPS, accelerometer, gyroscope and a camera enabling a low-cost, universal solution.
- ResQride combines a passive QR-based vehicle identity system with an optional smartphone crash-detection module.
- Working prototype built using React.js, React Native, Node.js, MySQL, Twilio, Google Maps and Razorpay.



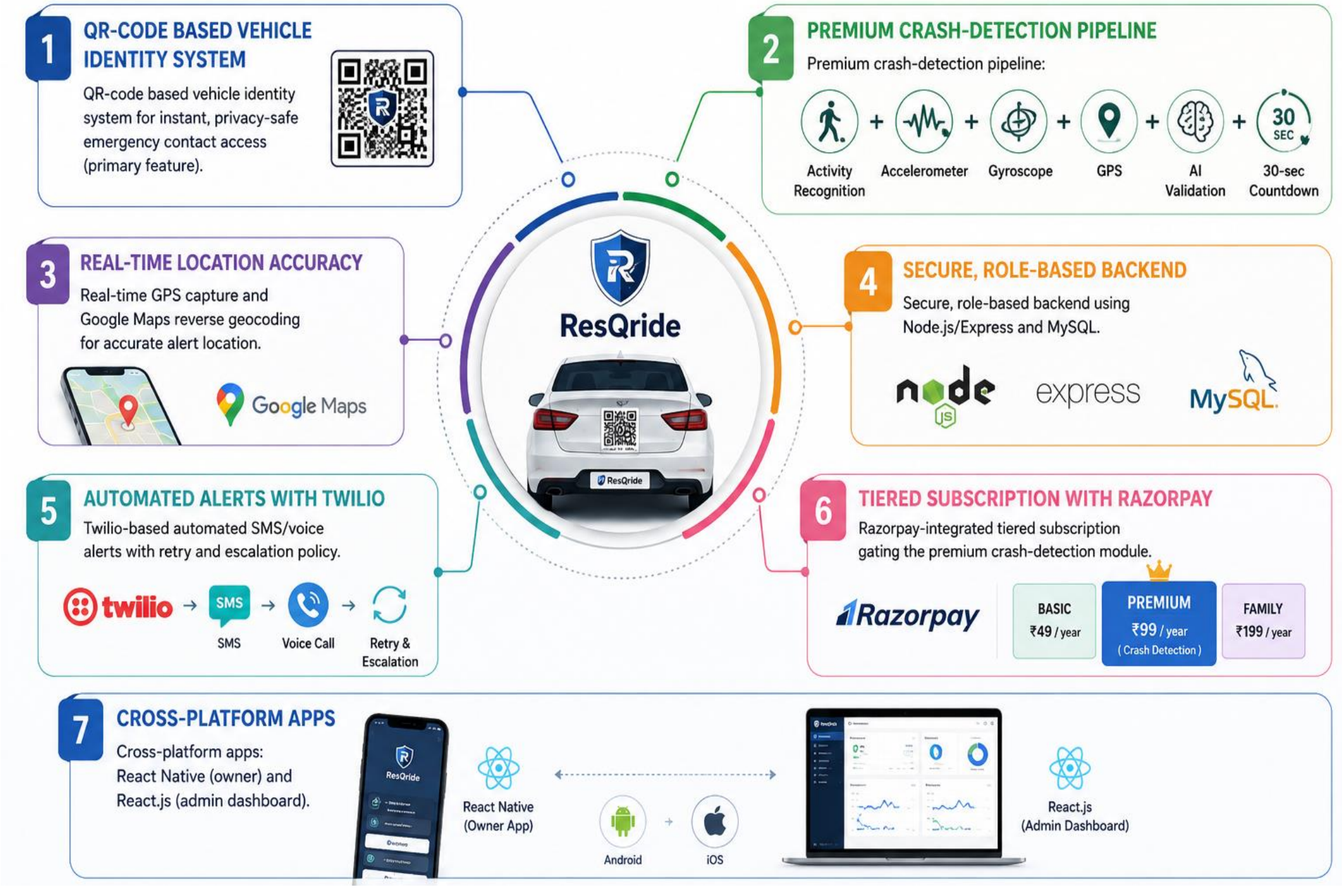
2. PROBLEM STATEMENT

- No low-cost way for bystanders or police to instantly reach a vehicle owner or their emergency contacts.
- High-end connected-car systems are bundled only with premium vehicles.
- Manual practices, such as searching a victim's phone, are slow, unreliable and privacy-invasive.
- Result: delayed notification, no automatic crash detection, and a privacy-vs-accessibility conflict.
- Emergency responders lose critical time identifying vehicle owners at accident scenes,
- Expensive hardware/subscription-based OEM systems — inaccessible for most vehicles, especially in Tier-2/3 cities.



3. OBJECTIVES

- QR-code based vehicle identity system for instant, privacy-safe emergency contact access (primary feature).
- Premium crash-detection pipeline: Activity Recognition + Accelerometer + Gyroscope + GPS + AI Validation + 30-sec Countdown.
- Real-time GPS capture and Google Maps reverse geocoding for accurate alert location.
- Secure, role-based backend using Node.js/Express and MySQL.
- Twilio-based automated SMS/voice alerts with retry and escalation policy.
- Razorpay-integrated tiered subscription gating the premium crash-detection module.
- Cross-platform apps: React Native (owner) and React.js (admin dashboard).



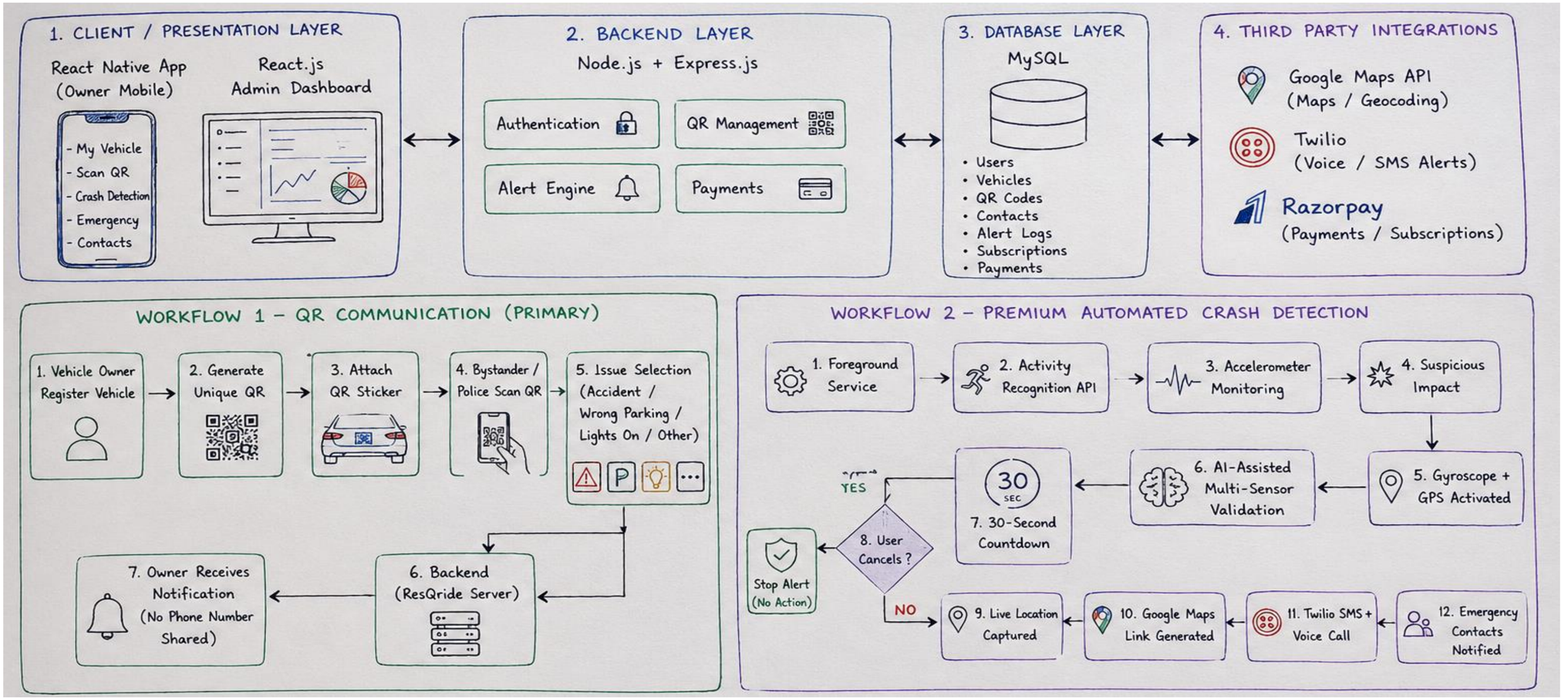
LITERATURE SURVEY

- IoT sensor-fusion crash detection validates impact-sensing as a reliable trigger — but needs dedicated hardware.
- Signal Magnitude Vector (SMV) is an established feature for distinguishing a crash from normal vibration.
- Combining GPS speed with accelerometer force improves collision-severity estimation.
- Alert-delivery speed matters more than detection accuracy alone for real-world outcomes.
- Gap addressed: ResQride delivers an equivalent safety layer purely via smartphone + QR tag — no dedicated hardware.
- Multi-sensor fusion (accelerometer + gyroscope + GPS) reduces false positives common in single-sensor crash detection



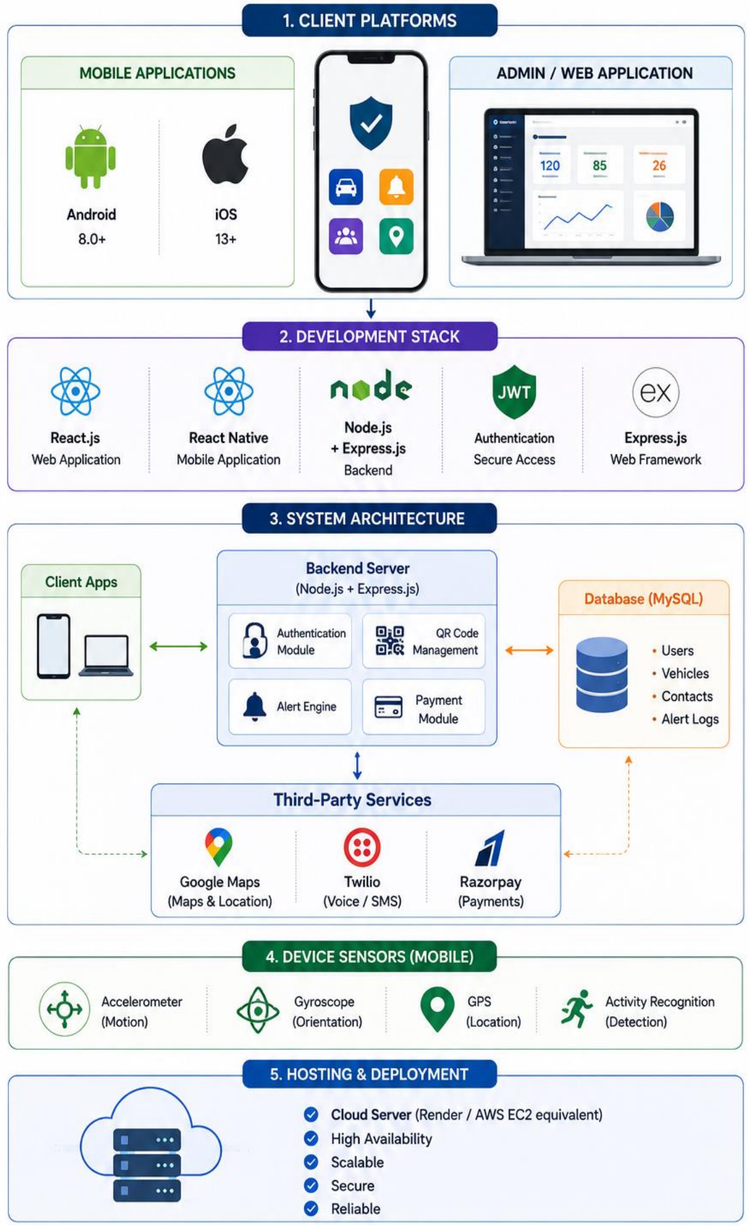
4. PROPOSED ARCHITECTURE & METHODOLOGY

- Four-tier architecture: Presentation (React.js / React Native) → API (Node.js/Express) → Data (MySQL) → Third-Party Integrations (Twilio, Google Maps, Razorpay).
- Two workflows: QR Communication (primary, always-available) and Premium Automated Crash Detection (below).



5. HARDWARE & SOFTWARE REQUIREMENTS

- Client OS: Android 8.0+ / iOS 13+ | Dev OS: Windows / macOS / Linux
- Stack: React.js, React Native, Node.js + Express, JWT authentication
- Database: MySQL (Users, Vehicles, Contacts, Alert Logs)
- Sensors: Accelerometer, Gyroscope, GPS, Activity Recognition
- Third-Party APIs: Twilio, Google Maps, Razorpay
- Hosting: Cloud server (Render / AWS EC2 equivalent)



6. EXPECTED DELIVERABLES

- Fully deployed QR-based emergency alert pipeline (primary feature).
- Premium crash-detection module with AI validation and 30-second countdown.
- Secure backend with an administrative web dashboard.
- Razorpay-integrated subscription flow.
- Complete project thesis with testing and evaluation report.
- Mobile app for both owners and rescuers – simple interface, no technical knowledge needed to use it
- Location sharing on alert – live GPS location sent automatically along with the alert, so help can reach faster.
- QR tag works without app install – bystander just scans and sends alert from any phone's browser, no download needed.
- SMS + call fallback – if internet is weak, alert still goes through via SMS/call using Twilio.
- Scalable cloud architecture – built to handle growing users without major backend changes.
- Cost-effective compared to OEM systems – no expensive hardware, works on any vehicle old or new

7. REFERENCES

- Ministry of Road Transport and Highways (MoRTH), Government of India, “Road Accidents in India – 2023,” Transport Research Wing, 2025.
- IIT Delhi, TRIPC, “Road Safety in India – Status Report 2023,” 2023.
- M. Andhare, O. Lomate, M. Tonde, G. Parwate, P. Bhute, “Emergency Information Sharing & Real-Time Communication via QR Codes,” Int. J. on Advanced Computer Engineering and Communication Technology, vol. 14, no. 1, 2025.
- “Accident QR System,” International Research Journal of Modernization in Engineering, Technology and Science (IRJMETS), 2024.
- LifeSaver HI, “Vehicle Sticker QR with Emergency Contact Features,” Product Documentation.
- “Real-Time IoT-based Vehicle Accident Detection and Emergency Response System using Multi-Sensor Fusion,” IEEE Conference Publication, 2025.
- A. Baladevan, “Signal-Based Crash Detection Using Accelerometer Data,” IJASIS, vol. 12, no. 3s, 2026.
- “Mobile Vehicle Crash Detection System,” IEEE Xplore.
- “Intelligent System for Vehicular Accident Detection and Notification,” IEEE Xplore.
- Twilio Inc., “Programmable SMS and Voice API Documentation,” Twilio Developer Docs.
- Google LLC, “Google Maps Platform — Geolocation and Geocoding API Documentation.”
- Android Developers, “Activity Recognition API Documentation,” Google Play Services.

THANK YOU !